ENGR-E 312 Lab 3: Romi Line Sensor Integration & Characterization

Jaiden Francisco Medina
Intelligent Systems Engineering, Luddy School of Informatics, Computing, and Engineering
Indiana University Bloomington
Bloomington, IN, USA
jfmedina@iu.edu

Brady Micheal Adams
Intelligent Systems Engineering, Luddy School of Informatics, Computing, and Engineering
Indiana University Bloomington
Bloomington, IN, USA
brmadam@iu.edu

***Abstract*—This project investigates radiation-induced fault behavior in a 4-bit-serial-in/serial-out CMOS shift register constructed from transistor-level, transmission-gate D flip-flops in LTSpice. Transient current pulses emulating heavy-ion strikes are injected at all critical storage nodes, including Q, Q̄, and internal master/slave latch nodes, to evaluate how charge deposition perturbs sequential state and propagates across pipeline stages. The analysis quantifies node-level susceptibility to Single-Event Upsets (SEU), Single-Event Transients (SET), and timing-related latch vulnerabilities. Sensitivity trends are correlated with electrical parameters such as node capacitance, drive strength, and logical/temporal masking. By mapping stage-by-stage vulnerability and identifying conditions under which disturbances persist or propagate, the study provides a structured methodology for SEE characterization in CMOS sequential circuits and informs the design of radiation-tolerant digital systems for space and aerospace applications.**

Keywords—radiation effects, single-event effects (SEE), single-event upset (SEU), single-event transient (SET), CMOS sequential logic, D flip-flop, shift register, transmission-gate flip-flop, transient current injection, LTSpice simulation, radiation tolerance, clock-phase sensitivity

I. Line Sensor Module installation and software functionality

A. Hardware Installation

The Romi reflectance sensor array consists of six infrared (IR) emitter–detector pairs mounted beneath the front of the robot. Each sensor emits IR light toward the surface and measures the intensity of the reflected signal. Light-colored surfaces reflect more IR radiation, while darker surfaces such as the black track line reflect less.

The sensor outputs were connected to the Romi 32U4 control board to allow the microcontroller to measure surface reflectance beneath each sensor.

B. Sensor Calibration

The sensors were calibrated prior to data collection by sweeping the array across both the white background and black line. This process establishes minimum and maximum reflectance values for each sensor, allowing readings to be normalized between light and dark surfaces.

C. Line Position Estimation Software

Calibrated sensor readings were processed using the QTRSensors library to estimate the position of the line relative to the center of the sensor array. The reported position was centered to produce a signed error value, where zero corresponds to a centered line, positive values indicate the line is to the right of center, and negative values indicate the line is to the left.

II. Line Sensor Accuracy Measurements

A. Experimental Procedure

The Romi Line sensor accuracy was evaluated by translating the Romi robot laterally relative to a stationary black line with increments of 0.25 inches across going 2 inches in the positive and negative directions. Four-line widths, 0.125 inches, 0.5 inches, 0.75 inches, and 1 inch, were tested in order to evaluate the effect of line width on the accuracy of the reflectance sensor array.

At each quarter inch displacement position, the robot was held stationary for approximately five seconds while the sensor-reported line position was recorded. Multiple readings were collected at each position and then averaged to reduce the influence of electrical noise and surface variation.

B. Error Calculation

The position of the line relative to the center of the sensor array was estimated using the QTRSensors centroid-based line position function. The reported

position was centered by subtracting the midpoint value to produce a signed displacement estimate.

The measurement at each displacement was computed as:

$$Error = Reported\ Displacement - Actual\ Displacement$$

Where the actual displacement was measured manually using a ruler and the reported displacement from the sensor position estimate.

C. *Error vs. Displacement Results*

The error values found for each tested line width were plotted as a function of actual line displacement, as shown in **Figure 1**. The expected ideal response corresponds to a horizontal line at zero error, indicating perfect agreement between the reported and actual line positions.

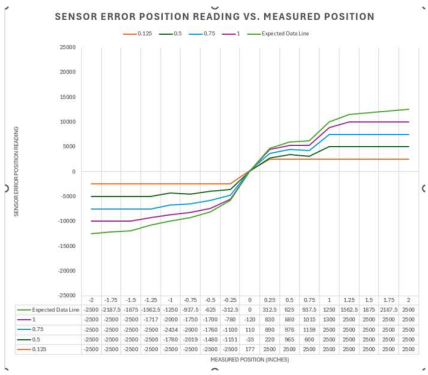


Figure 1: Graph of Sensor Error Position Reading vs. Measured Position

As shown in **Figure 1**, measurement accuracy varies significantly with line width. The line width of 0.125-inch line produced the largest deviation from the expected response. This has larger displacement from the center of measurement. This is likely due to insufficient sensor coverage of the narrow line, which resulted in reduced accuracy overall

The 1-inch line showed smoother sensor measurement but had visible bias. This was especially prevalent in larger offsets from the center measurement. This is likely due to multiple sensors detecting the line, which shifted the weighted position estimate and results in overestimation of displacement.

Comparatively, the 0.75 inch and 0.5 inch lines produced comparatively smaller deviations across the range of displacement. This indicates improvement in the agreement between the reported and actual line positions

III. Recommended Track Line Width

A. *Comparison of Measurement Error*

Comparison of the error curves for each line width indicates that the 0.5 inch line produces the most consistent and the most accurate displacement estimates across the test range of -2 to 2 inches from the center.

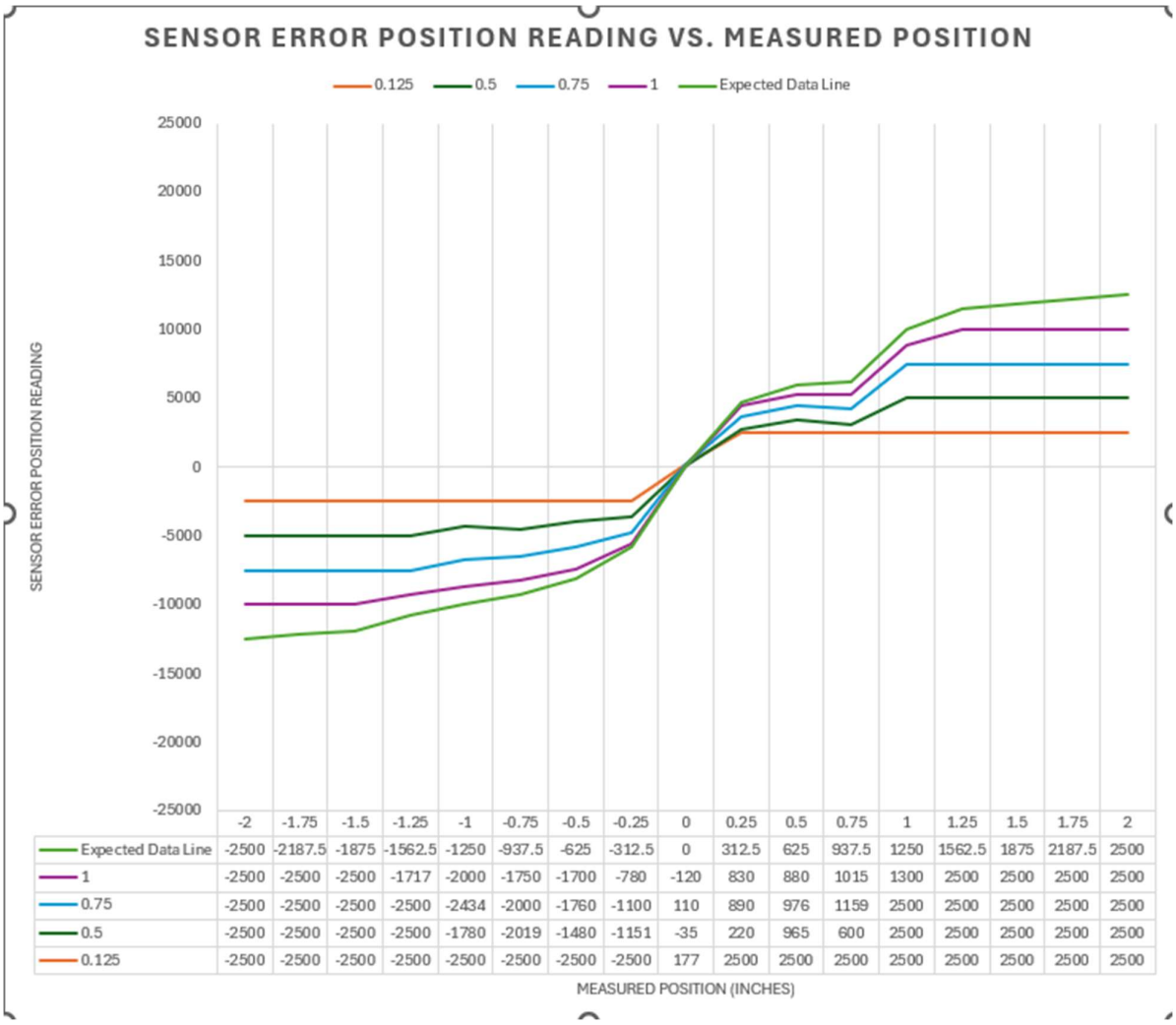
The narrower the lines resulted in increased variability an decreased accuracy due to incomplete sensor detection of the line surface. This is different than the wider lines, which introduced systematic bias as multiple sensors detected the line simultaneously

B. *Justifaction of Recommended Width*

Based on the measured trends, the 0.5 inch line width provides an optimal balance between sensor coverage and resolution, increasing accuracy. The width of 0.5 minimizes estimation error while maintaining sufficient contrast between the line and the background surface for reliable detection by the 6 individual sensors

C. *Final Recommendation*

Based on the observed measurement errors across the tested displacement range. Having a track line width of 0.5 inches is recommended for the Romi interface and sensor. This provided the most improved positional accuracy and reduced measurement variability in comparison to both the wider and narrower widths. Therefore, is expected to provide the most reliable performance.



	-2	-1.75	-1.5	-1.25	-1	-0.75	-0.5	-0.25	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2
Expected Data Line	-2500	-2187.5	-1875	-1562.5	-1250	-937.5	-625	-312.5	0	312.5	625	937.5	1250	1562.5	1875	2187.5	2500
1	-2500	-2500	-2500	-1717	-2000	-1750	-1700	-780	-120	830	880	1015	1300	2500	2500	2500	2500
0.75	-2500	-2500	-2500	-2500	-2434	-2000	-1760	-1100	110	890	976	1159	2500	2500	2500	2500	2500
0.5	-2500	-2500	-2500	-2500	-1780	-2019	-1480	-1151	-35	220	965	600	2500	2500	2500	2500	2500
0.125	-2500	-2500	-2500	-2500	-2500	-2500	-2500	-2500	177	2500	2500	2500	2500	2500	2500	2500	2500